

Qihang Jin

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EDUCATION

University of Science and Technology of China, Hefei, China Aug. 2024 – Jun. 2027
Master of Engineering in Electronic and Information Engineering

- GPA: 3.66/4.3; First-Class Graduate Scholarship, 2025
- Relevant Courses:** Computational Number Theory, Advanced Course of Artificial Intelligence, Computer Vision, Advanced Computer Networks, Principles of Neurobiology for Brain-Inspired Artificial Intelligence

Chang'an University, Xi'an, China Aug. 2018 – Jul. 2022
Bachelor of Engineering

PUBLICATION

Qinan Zhang, **Qihang Jin**. GridWM-Judge: Evaluating Vision-Language Model Judges in Grid Worlds via World Model Deficits. *ICLR 2026 Workshop on World Models*. 2026.

Qihang Jin, Enze Ge, Yuhang Xie, *et al.* Multimodal Representation Learning and Fusion. *arXiv*. Jun. 2025.
DOI: [10.48550/arXiv.2506.20494](https://doi.org/10.48550/arXiv.2506.20494)

Enze Ge, **Qihang Jin**, Yuhang Xie, *et al.* From Memory to Alignment: A Comprehensive Review of Large Language Model Optimization. *TechRxiv*. Oct. 2025.
DOI: [10.36227/techrxiv.176107630.07942950/v1](https://doi.org/10.36227/techrxiv.176107630.07942950/v1)

Qihang Jin, *et al.* Effect of Brain-Computer Interface on Limb Motor Function after Intracerebral Hemorrhage in Basal Ganglia and Its Rehabilitation Mechanism. *Under review*. Jun. 2025.

Jin Qihang, Cheng Zhaozhan. Adjustable Limiting and Fixing Device for Automated Machining. *CN 110480545 B*. Filed Aug. 23, 2019. Issued Jul. 9, 2021.

RESEARCH EXPERIENCE

Research Assistant, Columbia University — *Liam Paninski Lab* Dec. 2025 – present

- Neural signal analysis; IBL (International Brain Lab) data 3D reconstruction

HySSM-Pyramid: Learnable Hypergraph Scans for Multi-Scale Vision Jul. 2025 – present

- Designed HySSM (state-on-hyperedge SSMs) with $\mathcal{O}(N(d+h))$ complexity; verified Graph/Hypergraph degradation across multi-scale benchmarks.
- Built hypergraph builder, DAS-H scanner and HyperEdge-SSM stack with register anchors and fused SpMM.

J-ORA: Jacobi Orthogonal Rotation Adapter Jul. 2025 – present

- Proposed bilateral sparse Givens “sandwich” with Tiny Core (only 3.6K params/layer), Cayley-initialized rotations, and OER magnitude heads.
- Integrated with Triton kernels and LLaMA-Factory; delivered LoRA-vs-JORA benchmark suites.

Automotive Performance Data Processing | *MATLAB/GUI* Sept. 2021 – Jun. 2022

- Developed multi-interface application for automotive dynamics, fuel and braking KPI analysis; improved efficiency by 10%.

PROJECT EXPERIENCE

Integrated UAV Inspection Training Platform | *AirSim+UE4.27* Sept. 2024 – present

- Built UAV training platform with autonomous-flight and collision pipelines; reduced response latency 10% and raised efficiency 5%.

License Plate Recognition (CNN) | *Project Leader* Mar. 2020 – Aug. 2020

- 95%-accurate CNN-based system with OpenCV; robust to adverse lighting. **"Internet+" Bronze Award**.

SKILLS

Languages: Chinese (Native), English (Fluent, IELTS 6.5)

Technical: Python, C++, PyTorch, Scikit-learn, Pandas, NumPy